handwritten text extraction. But improving the success rate up from 90 per cent and adding more languages are definitely on the cards.

## Amazon Personalize

Amazon Personalize uses services like storage and analytics for a better user experience and for seamless development of ML models.

It is the most popular Amazon shopping platform, where recommendations are given for products based on the location and purchase history. It therefore gives a better user experience. The Amazon Personalization API has in-built service in a self-managed platform covering data inspection (history and profile), identification of metadata and features, selection of hyper parameters based on recommendations, training the ML models using these recommendations, optimisation of the ML model for better results, deployment of the ML models and sharing the recommendation results – all in an automated fashion.

Such personalised recommendations are really useful to build powerful user experience portals.

If you don't know much about building dynamic Web pages or about training ML models but want to develop an integrated ML model based user experience, then you can try out Amazon Personalize. With its public API facility, you can use it for integrated architecture with AWS applications or other cloud service providers (GCP, Azure). You can even use it for non-cloud application development/integration by sending input data through the Personalize API and receive the recommendations.

## Amazon SageMaker

Amazon SageMaker is an elastic, self-managed container instance. Data scientists or data analysts can use it to quickly spin-off instances to develop and deploy models in Jupyter Notebook without any cloud deployment and pipeline knowledge. It is also integrated automatically with Amazon CloudWatch monitoring and logging to enable serverless application management.

Data scientists and data analysts can use SageMaker to build the model and deploy it directly to a production-ready EC2 instance without the overhead of a DevOps pipeline. It is suitable for end-to-end machine learning model development, training and deployment on a serverless elastic platform.

Amazon SageMaker is a very popular and powerful service to build, train, manage and deploy ML models. We can use it for various ML activities with the many frameworks available inline such as:

- *SageMaker Canvas* can be used for ML predictions with its easy to navigate visual interface and drag and drop facility.
- *SageMaker Studio* can be used to build, train, manage and deploy ML models for data scientists to automate specific ML use cases.
- *SageMaker MLOps* enables deployment of ML models, and manages them at scale without worrying about infrastructure and configuration requirements.

## Amazon Cost Anomaly Detection

AWS Cost Anomaly Detection is a monitoring feature, which utilises advanced ML techniques to identify anomalous and suspicious spend behaviours as early as possible so you can avoid costly surprises. Based on the selected spend segments, Cost Anomaly Detection automatically determines patterns each day by adjusting for organic growth and seasonal trends. It triggers an alert when spend seems abnormal.

The different monitors are listed below.

- **AWS Services:** This monitor evaluates each of the services you use individually, allowing smaller anomalies to be detected. Anomaly thresholds are automatically adjusted based on your historical service spend patterns.
- **Linked Account:** This monitor evaluates the total spend for an individual linked account. This monitor can be helpful if your organisation defines teams (or products, services, and environments) by a linked account.
- **Cost Category:** This monitor evaluates the total spend for an individual cost category value. This monitor can be helpful if your organisation defines teams (or products, services, and environments) by using cost categories.
- **Cost Allocation Tag:** This monitor evaluates the total spend for an individual tag key-value pair. This monitor can be helpful if your organisation defines teams (or products, services, and environments) by using cost allocation tags.

## Amazon Kinesis

Amazon Kinesis is a managed, scalable, cloud based service that allows real-time processing of a large amount of streaming data per second. It is designed for real-time applications, and allows developers to take in any amount of data from several sources.

It is used to capture, store, and process data from large, distributed streams such as event logs and social media feeds. After processing the data, Kinesis distributes it to multiple consumers simultaneously.

The features of Amazon Kinesis are:

- Allows us to collect and analyse information in real-time like stock trade prices; otherwise, we need to wait for a data-out report.
- Developers can create a new stream, set its requirements, and start streaming data quickly.
- Can be integrated with Amazon